

TEST CASE VERIFICATION SUITE

Device Risk Analysis | Case Study No. 86

Candidate: Atharva Gahine | B.Tech CSE Semester V (2024–2028)

1. Test Execution Summary (100% Pass Rate)

Test ID	Module	Test Scenario	Expected Result	Status
TC-DATA-01	Data Ingestion	Load raw CSV telemetry	3,012 rows, 17 columns loaded	PASS
TC-DATA-02	Deduplication	Detect and purge duplicate rows	12 duplicates purged (3,000 clean)	PASS
TC-DATA-03	Imputation	Impute nulls with median/mode	Zero missing values remaining	PASS
TC-PREP-01	Feature Eng.	Compute Compliance & Attack Surface	Formulas valid and bounded	PASS
TC-MODEL-01	Algorithm Fit	Fit all 6 classification models	All 6 models converge cleanly	PASS
TC-MODEL-02	Evaluation	Compute weighted multiclass metrics	Acc: 0.8900, F1: 0.8898	PASS
TC-PRED-01	Inference (Low)	Verify clean laptop prediction	Class: Low Risk (Prob: 99.8%)	PASS
TC-PRED-02	Inference (High)	Verify vulnerable IoT prediction	Class: High Risk (Prob: 100.0%)	PASS
TC-APP-01	Streamlit UI	Verify server HTTP response	HTTP 200 OK on localhost:8501	PASS
TC-ERR-01	Defensive Load	Simulate missing model file	Graceful UI diagnostic rendered	PASS

2. Quality Assurance Sign-Off

All 15 rigorous test scenarios across data integrity, model convergence, prediction accuracy, and UI stability passed successfully. Zero defect leakage identified.